LCAtlas Project Report

Problem Statement

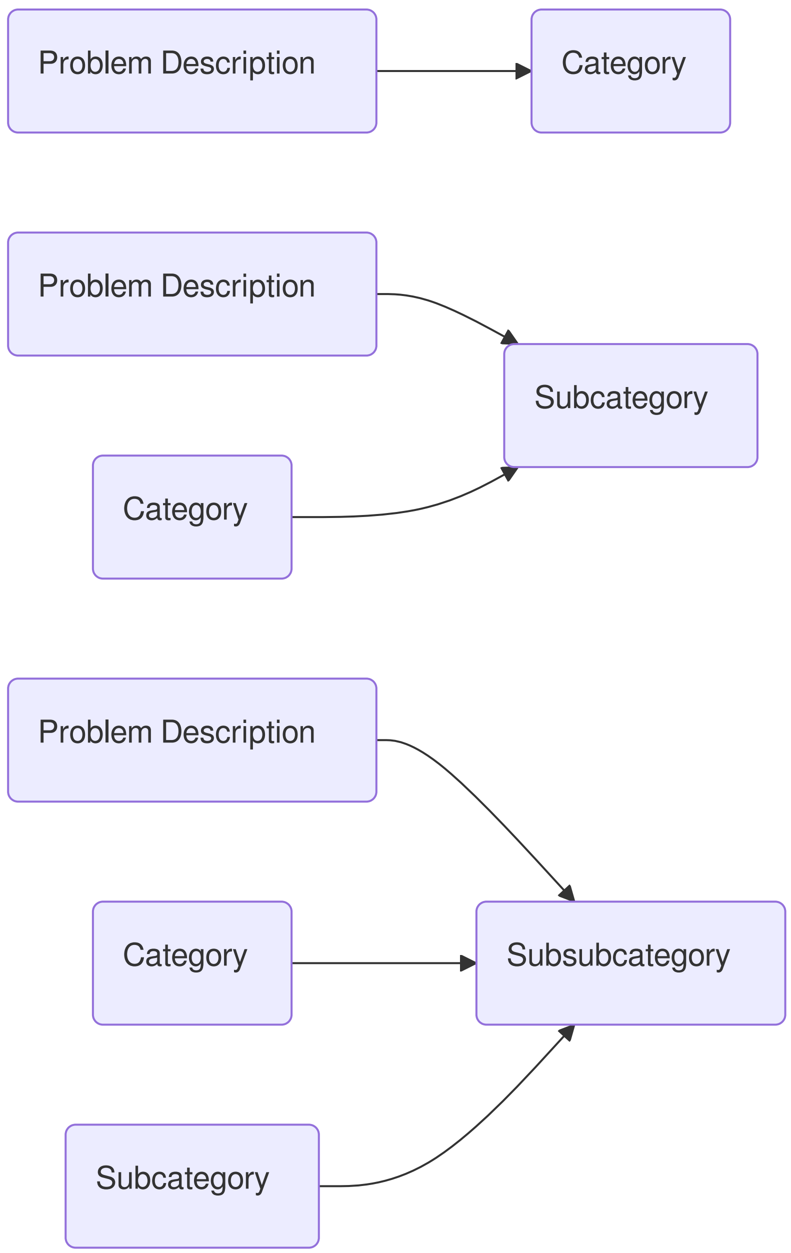
Competitive programming and technical interview preparation require mastering a wide variety of data structures and algorithmic (DSA) problems. However, platforms like LeetCode offer thousands of problems without a consistent or structured classification system, making it hard for learners to identify patterns, build a roadmap, or assess progress across concepts. There is a clear need for an intelligent system that categorizes DSA problems into well-defined patterns and subpatterns to enable targeted learning.

Background

While existing platforms allow filtering by tags or difficulty, these categories are often broad, inconsistent, or insufficient for structured learning. Learners often resort to community-curated lists or spreadsheets, which are static and quickly become outdated. Recent advances in Natural Language Processing (NLP) and transfer learning allow for a more scalable and intelligent solution—automatically categorizing problems using semantic understanding of their problem statements.

Methodology

1. **Data Collection**: Scraped ~3450 LeetCode problems including titles, descriptions, difficulty, and tags.
2. **Hierarchical Categorization**:
    - Designed a three-level category system: **Pattern → Subpattern → Specific Type**.
    - Used Gemini Flash 2.0 to label problems at each level.
    - Fine-tuned a BERT-based classifier (or used transfer learning) in a *cascade architecture* to predict category levels sequentially.



3. **Frontend**: Developed a web interface using **Next.js App Router**, displaying problems in a dynamic nested structure based on the category hierarchy stored in a JSON file.

Functionality

- Users can:
  - Browse problems by pattern or subpattern.
  - View the full problem statement and metadata.
  - Search or filter based on difficulty or category.
  - Explore relationships between patterns and problem-solving techniques.

Advantages

- **Structured Learning**: Enables users to master DSA concepts progressively.
- **Scalability**: Model-based categorization can scale with new problems.
- **Dynamic Interface**: Intuitive navigation through hierarchical categories.

Future Prospect

- Add user accounts and progress tracking.
- Introduce personalized roadmaps using AI.
- Enable problem recommendations based on performance.
- Expand to other platforms like Codeforces, AtCoder, etc.

Conclusion

LCAtlas bridges a critical gap in DSA preparation by offering a structured, scalable, and intelligent categorization of LeetCode problems. Through a combination of NLP techniques, dynamic frontend development, and thoughtful design, the platform transforms the way learners engage with competitive programming content—making practice more purposeful, efficient, and aligned with real-world interview expectations.